FIITJEE

JEE(Advanced)-2019

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

PART TEST – I

PAPER-1

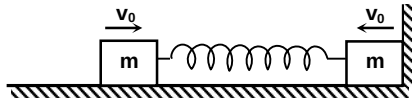
Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**A, B, C, D**	19.	**A, B, D**	37.	**B, C**
2.	**A, C**	20.	**B, C, D**	38.	**A, B**
3.	**A, C, D**	21.	**B, C, D**	39.	**A, B**
4.	**D**	22.	**A, B, C, D**	40.	**A, B**
5.	**A, C, D**	23.	**A, C**	41.	**A, B**
6.	**B, C, D**	24.	**A, B**	42.	**B, C**
7.	**D**	25.	**B**	43.	**C**
8.	**D**	26.	**D**	44.	**A**
9.	**B**	27.	**B**	45.	**D**
10.	**A**	28.	**C**	46.	**D**
11.	**00001.00**	29.	**00001.35**	47.	**00097.20**
12.	**00020.00**	30.	**00009.75**	48.	**00000.75**
13.	**00002.00**	31.	**00030.60**	49.	**00002.50**
14.	**00002.00**	32.	**00046.06**	50.	**00000.50**
15.	**00004.00**	33.	**00037.80**	51.	**00089.00**
16.	**00002.00**	34.	**00012.50**	52.	**–00012.50**
17.	**00007.00**	35.	**00012.25**	53.	**00000.50**
18.	**00003.00**	36.	**00303.90**	54.	**00000.40**

Physics

PART – I

SECTION – A

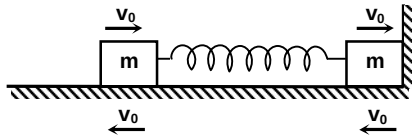
1. Immediately after 1st collision:
Since the situation is symmetric, both will come to rest simultaneously.
x_0 : maximum compression



$$\therefore \quad 2\times\frac{1}{2}mv_0^2=\frac{1}{2}kx_0^2$$

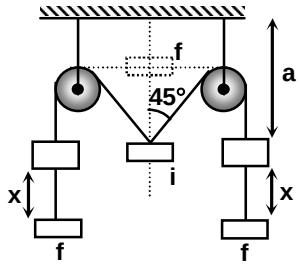
$$\therefore \quad x_0=v_0\sqrt{\frac{2m}{K}}$$

Just before 2nd collision:
Finally:



$$\therefore \quad \Delta\vec{P}=4mv_0(\leftarrow)$$

2. Applying work energy theorem from initial to final state (i to f)



$$2Mgx - mgy = 0$$

$$\Rightarrow \quad 2Mx = mg \qquad \ldots(i)$$

Keeping length of the string constant

$$\Rightarrow \quad \sqrt{2}y + 0 = y + x + a$$

$$y(\sqrt{2}-1) = x \qquad \ldots(ii)$$

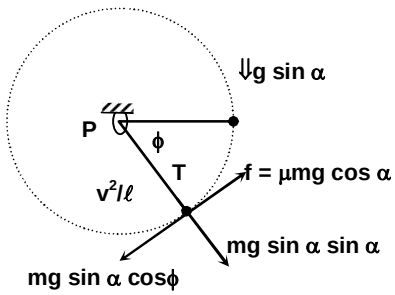
From (i) and (ii)

$$\Rightarrow \quad 2My(\sqrt{2}-1) = my$$

$$\Rightarrow \quad m = 2M(\sqrt{2}-1)$$

In the final state instantaneous acceleration of m has to be g as there will be only one force (That's mg). and the acceleration of masses M are zero at the same moment .

3.



$$T - mg\sin\alpha\sin\phi = \frac{mv^2}{\ell} \qquad \ldots(i)$$

$$mg\sin\alpha\cos\phi - \mu\, mg\cos\alpha = m\frac{dv}{dt} \qquad \ldots(ii)$$

work energy theorem

$$(mg\sin\alpha)\,\ell\sin\alpha - (\mu mg\cos\alpha)\ell\,\phi = \frac{1}{2}mv^2 \qquad \ldots(iii)$$

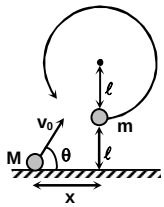
From (i) and (iii)

$$\Rightarrow T - mg\sin\alpha\sin\phi = \alpha mg\sin\alpha\sin\phi - 2(\mu mg\cos\alpha)\,\phi$$

$$\Rightarrow T = mg(3\sin\alpha\sin\phi - 2\mu\,\phi\cos\alpha)$$

$$\frac{dT}{d\phi} = 0 = 3\sin\alpha\cos\phi - 2\mu\cos\alpha$$

4.



$$Mv_0\cos\theta + 0 = Mv + mv'$$

$$e = 1 = \frac{v'-v}{v_0\cos\theta}$$

$$v' = \frac{2Mv_0\cos\theta}{M+m} \geq \sqrt{5g\ell}$$

Also, x = R/2 and ℓ = H

Solving $\therefore\ x \geq \frac{3\sqrt{10}}{4}$

$x \geq 2.37$ m

5. $F_{ext} = kx + mg$ (x : extension at any t)

$\therefore\ W_{ext} = \int_0^{x_0} (kx + mg)dx$, where $kx_0 = mg$

Releasing, total work done = $W_{ext} + mg(h)$, where h : natural length

6. Underestimate the work done against friction, and compare it with the initial gravitational potential energy of the eraser.

First of all, we investigate whether the rubber eraser will start moving at all. It will do so provided that mg sin α>μmg cos α, i.e. μ< tan α = tan 45◦ = 1. This is clearly the case, since μ = 0.6. So, the eraser will start moving. The trouble is that the determination of how the normal force acting on the eraser varies with position is difficult. A calculation of the work done against friction can be carried out, to any given degree of accuracy, only by using a computer. The trouble is that the determination of how the normal force acting on the eraser varies with position is difficult. However, it is certain that the frictional force is always larger than its initial value of μmg cos α = μmg cos 45◦ This is because, after the initial release, the angle with the horizontal made by the slope on which the eraser moves decreases, and, in addition, the track has to provide a centripetal force for the moving eraser. The path to reach the lowest point of the track would be one-eighth of a circle, with a length of Rπ/4. The work done against friction can now be underestimated by taking the normal force as if it always had its initial value:

$$\left|W_f\right| > \left|W_{und}\right| = \frac{\mu mg}{\sqrt{2}}\frac{\pi R}{4} \approx 0.333 mgR$$

The gravitational potential energy difference between the initial position and the bottom of the track is $\Delta E_p = mgR(1 - \cos\alpha) = mgR(1 - \cos 45^\circ) \approx 0.293mgR$.

It can be seen that $|W_f| > \Delta Ep$, i.e. the work to be done against friction is clearly larger than what can be provided by the gravitational potential energy. So, the rubber eraser cannot reach the very lowest part of the track.

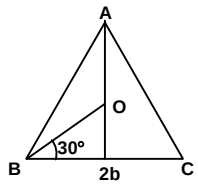
7-8. The pipes just below and just above the sheet which is moving will move the other pipes and the other pipes will remain in the original place. If the sheet is moved by distance greater then 5 cm the centre of gravity of the system would fall outside the base and the system will collapse.

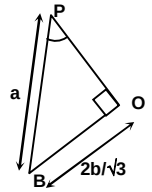
9-10. $\cos 30° = \frac{b}{(BO)} = \frac{\sqrt{3}}{2} \Rightarrow BO = \frac{2b}{\sqrt{3}}$

Let P be particle clearly,

$3T_i \cos\theta = mg$

$T_i = \frac{mg}{3}\sec\theta = \frac{mga}{\sqrt{3(3a^2 - 4b^2)}}$





SECTION – D

11. Imagine the apex of the cone being depressed by Δh, and the radius of its base consequently increasing by Δr, i.e. the perimeter of the base circle increases by $2\pi\Delta r$. Then the total work done by external forces would be:

$w\Delta h - F(2\pi\Delta r) = 0$

$\Rightarrow \quad F = \frac{w}{2\pi}\frac{\Delta h}{\Delta r}$

However, the changes in height and base circle radius are not independent, the connection between them being determined by the fixed length ℓ of one of the straight edges of the sheet. Using Pythagoras's theorem:

$(r + \Delta r)^2 + (h - \Delta h)^2 = \ell^2 = r^2 + h^2$

from which we have

$\frac{\Delta h}{\Delta r} = \frac{2r + \Delta r}{2h - \Delta h} \approx \frac{r}{h}$

So the force in question has magnitude

$F = \frac{w}{2\pi}\frac{r}{h} = 1$

12. The weight of the chain is balanced by the vertical components of the reaction forces at the two suspension points. The magnitude of each of these components must be $F_0 = \rho Lg/2$, where ρ is the mass of the chain per unit length, and L = 40 cm. If the chain made an angle of θ with the vertical at a suspension point, the horizontal component of the chain's tension would be $F_0\tan\theta$. Because, in practice, $\theta = 45^\circ$, the horizontal component is the same as the vertical one; it is also constant along the chain, since there are horizontal forces acting on the chain only at its ends.

At the chain's lowest point, the tension is purely horizontal, and, as just shown, its magnitude is F_0. So, around the lowest point, consider a small piece of the chain that subtends an angle 2θ at the centre of the osculating circle, whose radius is the value r_1 we seek. The length of the piece is $2\theta r_1$ and it is pulled down by a gravitational force $2\theta r_1\rho g$. This force is balanced by the upward net force of $2F_0\sin\theta$ due to the tension in the chain. In the limit of small angles, when $\sin\theta \approx \theta$, the equilibrium equation

$2\theta r_1\rho g = 2F_0 \sin\theta$

leads to

$F_0 = \rho g r_1$.

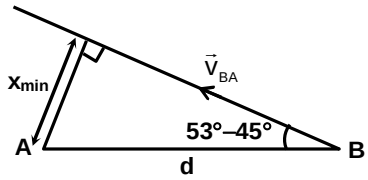
Since $F_0 = \rho Lg/2$, we have the simple result that $r_1 = L/2 = 20$ cm.

13. **Method 1:** Even though the normal force with the floor does no physical work on the person (energy transfers occur internally from the muscles), one can write a mathematically correct expression as though the contact force with the floor does change the person's energy. Hence, we can treat the center of mass motion as Wnet = ΔKcm. Here, we are treating it as though only gravity and the floor act onthe person, and since the person is in contact with the floor until the center of mass is a position h/2 from the ground, then $F_{floor}\frac{h}{4} + (mg)\left(\frac{-h}{2}\right) = 0$

as the student starts and ends this motion at rest. Therefore, $F_{floor} = 2mg$.

Method 2: Since the feet come off the floor at a position h/2 from the ground, then the person accelerates upward from h/4 to h/2 with the same magnitude as the acceleration from h/2 to 3h/4 by symmetry. Therefore, the net force on the student going upward is the same in magnitude as the student in free fall from h/2 to 3h/4. Hence, while in contact with the floor, using Newton's second law, $F_{floor} - mg = mg \Rightarrow F_{floor} = 2mg$.

14. $x_{min} = d \sin(53°-45°) = \dfrac{d}{5\sqrt{2}} = 2$

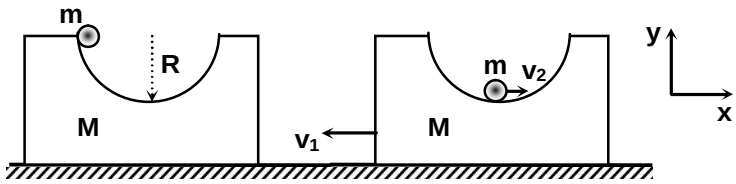


15. Momentum conservation along x axis

$0 = mv_2 - Mv_1$

Work energy theorem

$mgR = \dfrac{1}{2}mv_2^2 + \dfrac{1}{2}Mv_1^2$



from (i) and (ii)

$\Rightarrow 2mgR = mv_2^2 + \dfrac{m^2}{M}v_2^2$

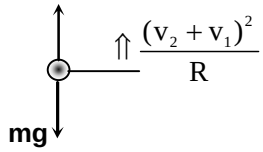
$\sqrt{\dfrac{2gRM}{M+m}} = v_2$ and $v_1 = \dfrac{m}{M}\sqrt{\dfrac{2gRM}{M+m}}$

Making FBD of particle with respect wedge at lowest point.

$\dfrac{7mg}{2} - mg = \dfrac{m(v_1+v_2)^2}{R}$

Substituting the values, we get : $\dfrac{M}{m} = 4$

N = (7mg/2)



16. Let us call h the height above the ground at which the projectile explodes and v_{jy} (j=1, 2, 3) the y-component of the velocity at the moment of the explosion for the three fragments. I will label with 1 the fragment which lands after t seconds.

Since the landing time depends only on the y-component of the velocity at the moment of the explosion, we have: $v_{3y} = v_{2y}$. We then need only the equations for fragments 1 and 2, stating that at landing time the y-coordinate will be equal to 0:

$h + v_{1y}t - \dfrac{gt^2}{2} = 0$

$h + v_{2y}t - \dfrac{g(2t)^2}{2} = 0$

Since the masses of the three fragments are equal, and the y-component of the projectile momentum at the moment of the explosion is equal to 0 (it explodes at the top of the trajectory), the momentum conservation law reads:

$v_{1y} + v_{2y} + v_{3y} = 0$.

The solution of the above equations is then:

$v_{1y} = \dfrac{-3}{4}gt$

$v_{2y} = v_{3y} = \dfrac{3}{8}gt$

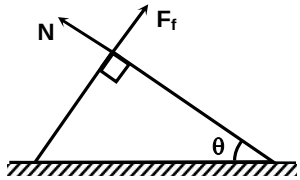
The height above the ground at which the projectile explodes is then: $h = 5gt^2/4$

17. Tension T = weigth of cat (since cat is at rest).

$\therefore$ Net downward force on rod = T + 4g

$a_{rod} = \dfrac{3g+4g}{4} = 7g/4\ m/s^2$

18. Let M_l be the mass of the left stick, and let M_r be the mass of the right stick. Then $M_l/M_r = \tan\theta$. Let N and F_f be the normal and friction forces between the sticks. F_f has a maximum value of μN. Balancing the torques on the left stick (around the contact point with the ground) gives $N = (M_l g/2)\sin\theta$. Balancing the torques on the right stick (around the contact point with the ground) gives $F_f = (M_r g/2)\cos\theta$. The condition $F_f \le \mu N$ is therefore



$$M_r \cos\theta \le \mu M_1 \sin\theta \Rightarrow \tan^2\theta \ge \frac{1}{\mu}$$

where we have used $M_l/M_r = \tan\theta$.

Chemistry

PART – II

SECTION – A

19. $\underset{4-x}{PCl_5(g)} \rightleftharpoons \underset{x}{PCl_3(g)} + \underset{x}{Cl_2(g)}$

Total moles at equilibrium = 4 + x + 2 = 6 + x

$$n = \frac{PV}{RT}$$

$$6 + x = \frac{6.568 \times 80}{0.0821 \times 800}$$

$$6 + x = 8, x = 2$$

$$[PCl_5] = \frac{2}{80}, [PCl_3] = \frac{2}{80}. [Cl_2] = \frac{2}{80}$$

$$K_c = \frac{\frac{2}{80} \times \frac{2}{80}}{\frac{2}{80}} = 0.025$$

$$K_P = K_c (RT)^{\Delta n}$$

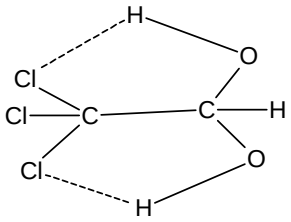
= 0.025 × 0.0821 × 800

= 1.642 atm

20. Bond angle of NH_3 > Bond angle of NF_3

21. $B_2O_3 + P_2O_5 \longrightarrow 2BPO_4$

22.



Chloral hydrate

23. $2KMnO_4 + 5K_2C_2O_4 + 8H_2SO_4 \longrightarrow 2MnSO_4 + 10CO_2 + 6K_2SO_4 + 8H_2O$

24. $Zn(OH)_2 + 2NaOH \longrightarrow \underset{(Soluble)}{Na_2ZnO_2} + 2H_2O$

$Al(OH)_3 + NaOH \longrightarrow \underset{(Soluble)}{NaAlO_2} + 2H_2O$

25. $\left[Ag^+\right] = \sqrt[3]{\dfrac{K_{sp}}{\left[PO_4^3\right]}}$

$= \sqrt[3]{\dfrac{2.7\times10^{-18}}{0.1}}$

$= 3 \times 10^{-6}$ M

26. $\left[Cl^-\right] = \dfrac{K_{sp}AgCl}{\left[Ag^+\right]}$

$= \dfrac{1.2\times10^{-10}}{3\times10^{-6}}$

$= 4\times10^{-5}$

27. $K_p = P_{NH_3} \times P_{H_2S}$

$0.36 = P_{NH_3} \times P_{H_2S}$

$P_{NH_3} = P_{H_2S} = 0.6$ atm

$P_{NH_3} + P_{H_2S} = 0.6 + 0.6 = 1.2$ atm

28. (i) K_P is constant at a given temperature.
(ii) Addition of inert gas at constant volume does not affect equilibrium.

29. $K_c = 0.36, Q_c = \dfrac{1.8\times1.8}{1.8\times1.8} = 1$

$\underset{1.8+x}{SO_2(g)} + \underset{1.8+x}{NO_2(g)} \rightleftharpoons \underset{1.8-x}{SO_3(g)} + \underset{1.8-x}{NO(g)}$

$\therefore K_c = \dfrac{(1.8-x)^2}{(1.8+x)^2}$

$0.36 = \dfrac{(1.8-x)^2}{(1.8+x)^2}$, $\qquad 0.6 = \dfrac{1.8-x}{1.8+x}$

$\therefore x = 0.45$

Number of mole of SO_3 = 1.8 – 0.45 = 1.35

30.

	H_3PO_4	+ NaOH	$\longrightarrow NaH_2PO_4$	+ H_2O
M.mole	2	4	0	
	0	2	2	

	NaH_2PO_4	+ NaOH	$\longrightarrow Na_2HPO_4$	+ H_2O
M.mole	2	2	0	
	0	0	2	

pH of Na_2HPO_4 = $\frac{pK_{a_2} + pK_{a_3}}{2}$

$= \frac{7.2 + 12.3}{2} = 9.75$

31. $KE = 13.6 \times \frac{Z^2}{n^2}$

$= 13.6 \times \frac{3^2}{2^2} = 30.6 \text{ eV}$

32. $k = \frac{2.303}{t} \log \frac{[A]_o}{[A]}$

$t = \frac{2.303 \times 6.93}{0.693} \log \frac{0.1}{0.001}$

$t = 46.06$

33. $\frac{E_{a_1}}{T_1} = \frac{E_{a_2}}{T_2}$

$\frac{42}{500} = \frac{E_{a_2}}{450}$

$E_{a_2} = 37.8 \text{ kJ}$

$\therefore x = 37.80$

34. $[A] = \frac{[A]_o}{2^3} = 0.125[A]_o$

$\therefore \%$ of A left $= \frac{0.125[A]_o}{[A]_o} \times 100 = 12.50$

35. $K_2Cr_2O_7 + 4H_2SO_4 + 3H_2C_2O_4 \longrightarrow 6CO_2 + 7H_2O + Cr_2(SO_4)_3 + K_2SO_4$

134.4 L of CO_2 produced by $K_2Cr_2O_7$ = 294 g

$\therefore$ 5.6 L of CO_2 produced by $K_2Cr_2O_7 = \frac{294}{134.4} \times 5.6$

= 12.25 g

36. $\frac{1}{\lambda} = R(Z)^2 \left[\frac{1}{1^2} - \frac{1}{2^2}\right]$

$\frac{1}{\lambda} = R \times 4 \left[\frac{3}{4}\right]$

$\lambda = \frac{1}{R} \times \frac{1}{3} = 911.7 \times \frac{1}{3} \text{ Å}$

$\lambda = 303.90 \text{ Å}$

Mathematics

PART – III

SECTION – A

37. $\ln\beta = \lim\limits_{n\to\infty}\frac{1}{n^2}[\ln 1 + (\ln 1 + \ln 2) + (\ln 1 + \ln 2 + \ln 3) + ...] - \alpha \ln n$

$= \lim\limits_{n\to\infty}\frac{n\ln\frac{1}{n} + (n-1)\ln\frac{2}{n} + ... + \ln\frac{n}{n} + \frac{n(n+1)}{2}\ln n}{n^2} - \alpha\ln n = \lim\limits_{n\to\infty}\frac{n+1-2n\alpha}{2n}\ln n + \frac{1}{n}\sum\limits_{r=1}^{n}\frac{n+1-r}{n}\ln\frac{r}{n}$

$\Rightarrow \alpha = \frac{1}{2}$ and $\ln\beta = \int\limits_0^1 (1-x)\ln x\,dx = -\frac{3}{4}$

38. Let p = abc and q = [a] [b] [c]. So, q is an integer and $p = \sqrt{\frac{60}{q}}$

$q \le \min\{[a]bc, a[b]c, ab[c]\} = 3 \Rightarrow q \in \{1, 2, 3\}$
If $q = 1 \Rightarrow ab[c] < 4$ but $ab[c] = 5 \Rightarrow$ no solution
If q = 3, as $p \ge \max\{[a]bc, a[b]c, ab[c]\} = 5$ and $p = \sqrt{\frac{60}{q}} \Rightarrow$ no solution
$\therefore$ q = 2 and $p = \sqrt{30}$. Since q is a product of 3 positive integer, we have 3 cases
If [a] = 2, then $a = \frac{2}{5}\sqrt{30} = \sqrt{\frac{120}{9}} > 3 \Rightarrow$ no solution
If [b] = 2, then solution is $\left(\frac{\sqrt{30}}{3}, \frac{\sqrt{30}}{2}, \frac{\sqrt{30}}{5}\right)$
If [c] = 2, then solution is $\left(\frac{\sqrt{30}}{3}, \frac{\sqrt{30}}{4}, \frac{2\sqrt{30}}{5}\right)$

39. Let f′(1) = a and f″(1) = b, putting x = 1 in the given equation we get $b = a^2$
Differentiating the equation and putting x = 1 gives $2ab + 8 = ab + 2ab \Rightarrow ab = 8$
So, a = 2, b = 4

40. $I_n = \int\limits_{-\pi}^{0}\frac{\sin nx}{(1+2^x)\sin x}dx + \int\limits_0^{\pi}\frac{\sin nx}{(1+2^x)\sin x}dx$

$I_n = \int\limits_0^{\pi}\frac{\sin nx}{(1+2^{-x})\sin x}dx + \int\limits_0^{\pi}\frac{\sin nx}{(1+2^x)\sin x}dx = \int\limits_0^{\pi}\frac{\sin nx}{\sin x}dx$ [Putting –x for x in 1st integral]

$I_{n+2} - I_n = \int\limits_0^{\pi}\frac{\sin(n+2)n - \sin nx}{\sin x}dn = 2\int\limits_0^{\pi}\cos(n+1)x\,dx = 0$ if $n \in N$
$\Rightarrow$ If n is even $I_n = I_0 = 0$, and if n is odd $I_n = I_1 = \pi$

41. Using LH-Rule twice, we get $a = b = \pm 2$

42. $\lim\limits_{x\to 1}\frac{(\ln(2-x))^2}{x^2+ax+b} = 1 \Rightarrow 1 + a + b = 0$

Using LH-Rule, we have $\lim_{x\to 1}\frac{2\ln(2-x)}{(x-2)(2x+a)}=1 \Rightarrow a+2=0 \Rightarrow a=-2,\ b=1$

43. $\frac{f(x)}{1+x^2}=1+\int_0^x\frac{f(t)}{1+t^2}dt \Rightarrow \frac{f'(x)(1+x^2)-f(x)2x}{(1+x^2)^2}=\frac{f(x)}{1+x^2} \Rightarrow f'(x)-\frac{2xf(x)}{1+x^2}=f(x)$

$\Rightarrow \frac{f'(x)}{f(x)}=1+\frac{2x}{1+x^2} \Rightarrow \ln f(x)=x+\ln(1+x^2)+\ln c \Rightarrow f(x)=ce^x(1+x^2)$

44. Given equation implies $\frac{xf'(g(x))g'(x)}{f(g(x))}=\frac{g'(f(x))f'(x)}{g(f(x))}$

$\Rightarrow \frac{d}{dx}(\ln g(f(x)))=x\frac{d}{dx}(\ln f(g(x)))=-2x$ (as $f(g(x))=e^{-2x}$),

$\Rightarrow g(f(x))=Ae^{-x^2}$ and $g(f(0))=1 \Rightarrow A=1$

45. Using integration by parts repeatedly

$$\int_1^\infty\frac{(\ln x)^{2018}}{x^{2018}}dx=\left[\frac{(\ln x)^{2018}}{-2017x^{2017}}\right]_1^\infty-\int_1^\infty\frac{2018(\ln x)^{2017}}{-2017x^{2018}}dx=\frac{2018}{2017}\int_1^\infty\frac{(\ln x)^{2017}}{x^{2018}}dx$$

$$=\ldots..=\frac{2018!}{(2017)^{2018}}\int_1^\infty\frac{dx}{x^{2018}}=\frac{2018!}{(2017)^{2019}}$$

46. $I=2\int_0^1\frac{2x^{332}+x^{998}}{1+x^{666}}dx$, (as $\frac{4x^{1664}\sin x^{691}}{1+x^{666}}$ is an odd function and rest of the integrand is even)

Put $x=t^{\frac{1}{333}} \Rightarrow dx=\frac{1}{333}t^{-\frac{332}{333}}dt$

$$I=\frac{2}{333}\int_0^1\frac{2t^{\frac{332}{333}}+t^{\frac{998}{333}}}{1+t^2}t^{\frac{-332}{333}}dt=\frac{2}{333}\int_0^1\frac{2+t^2}{1+t^2}dt=\frac{2}{333}\left(1+\frac{\pi}{4}\right)$$

SECTION – D

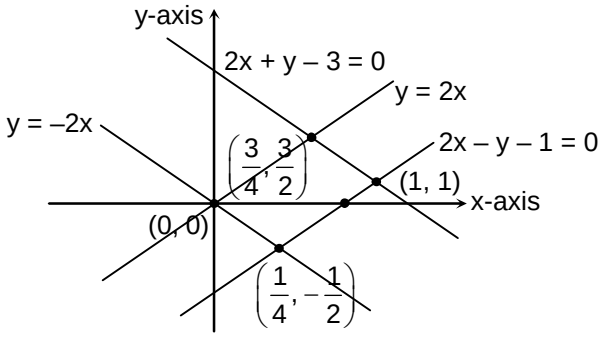
47. $\int_0^1 f(x)f'(x)dx=\left[\frac{(f(x))^2}{2}\right]_0^1=0 \Rightarrow f(0)=\pm f(1)$

$\int_0^1(f(x))^2f'(x)dx=\left[\frac{(f(x))^3}{3}\right]_0^1=18 \Rightarrow f(1)=3$ and $f(0)=-3$

48. Lines with slope ± 2 and passing through (0, 0) and (1, 1) are $y=\pm 2x$ and $y-1=\pm 2(x-1)$ respectively.

By the Mean Value theorem, no point of form (x, f(x)) lies outside this parallelogram. However, we can construct functions which satisfy the given condition and are arbitrarily close to the sides of the parallelogram.

So, b – a = area of parallelogram = $\frac{3}{4}$



49. Let $S_n = \int_{\frac{1}{2^n}}^{\frac{1}{2^{n-1}}} f(x)dx$, then $S_n = \frac{1}{3}\int_{\frac{1}{2^n}}^{\frac{1}{2^{n-1}}} f(2x)dx = \frac{S_{n-1}}{6} = ... = \frac{S_0}{6^n}$

$$\int_0^1 f(x)dx = \lim_{n\to\infty}\left(\int_{\frac{1}{2^n}}^{\frac{1}{2^{n-1}}} f(x)dx + \int_{\frac{1}{2^{n-1}}}^{\frac{1}{2^{n-2}}} f(x)dx + ... + \int_{\frac{1}{2}}^{1} f(x)\right) = \frac{S_0}{5} = 1 \Rightarrow S_0 = 5$$

50. $f(1-x) = 1 - f(x) \Rightarrow f^k(1-x) = 1 - f^k(x)\ \forall\ k \in N$

$$\int_0^1 f^{2004}(x)dx = \int_0^1 \left(1 - f^{2004}(1-x)\right)dx = 1 - \int_0^1 f^{2004}(1-x)dx = 1 - \int_0^1 f^{2004}(x)dx \text{ (Put } 1-x \text{ for } x)$$

51. If x_1, x_2, x_3 and x_4 are the root of p then $\frac{p'(7)}{p(7)} = \sum \frac{1}{7-x_i} = 0$ (as 7 is not a root of p(x))

$$\Rightarrow \frac{1}{7-1} + \frac{1}{7-3} + \frac{1}{7-5} + \frac{1}{7-x_4} = 0 \Rightarrow x_4 = \frac{89}{11}$$

52. $L = \lim_{x\to\infty} x^p\left((x+1)^{1/3} + (x-1)^{1/3} - 2x^{1/3}\right) = \lim_{t\to 0} t^{-p}\left(\left(\frac{1}{t}+1\right)^{1/3} + \left(\frac{1}{t}-1\right)^{1/3} - 2\left(\frac{1}{t}\right)^{1/3}\right)$, (putting $x = \frac{1}{t}$)

$= \lim_{t\to 0} t^{-p-\frac{1}{3}}\left((1+t)^{1/3} + (1-t)^{1/3} - 2\right) = -\frac{2}{9}t^{-p+\frac{5}{3}}$ (using expansion for $(1+x)^{1/n}$)

$\therefore\ p = \frac{5}{3}, L = -\frac{2}{9}$

53. $\frac{d}{dt}\int_{-\infty}^{\infty} e^{tx} f(x)dx = \int_{-\infty}^{\infty} x e^{tx} f(x)dx = \frac{d}{dt}\left(\sin^{-1}\left(t - \sqrt{\frac{1}{2}}\right)\right) = \frac{1}{\sqrt{1-\left(t-\frac{1}{\sqrt{2}}\right)^2}}$

Putting t = 0, $\int_{-\infty}^{\infty} x f(x)dx = \sqrt{2}$

54. The tangent at (x, f(x)) meets the y-axis at (0, f(x) –1)

$\therefore\ f'(x) = \frac{1}{x} \Rightarrow f(x) = \ln x + c$ and $f(1) = 0 \Rightarrow c = 0$

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